

Chapter 6

3. Let $\phi(x) = \sqrt{x}$

$$\phi(xy) = \sqrt{xy} = \sqrt{x}\sqrt{y} = \phi(x)\phi(y)$$

ϕ preserves group operation

one to one Let $\phi(x) = \phi(y)$. This means $\sqrt{x} = \sqrt{y}$.

Squaring both sides gives $x = y$.

Thus ϕ is injective

onto: Let $z \in \mathbb{R}^+$. Find x such that $\phi(x) = z$

This means $\sqrt{x} = z$. Squaring both sides give
 $x = z^2$

Since z is a positive real number, $z^2 \in \mathbb{R}^+$.

Thus ϕ is surjective.

$\therefore$ Since the operation validates these steps,

$\phi(x) = \sqrt{x}$ is an automorphism of $\mathbb{R}^+$

4. $U(8) = \{1, 3, 5, 7\}$

$$U(10) = \{1, 3, 7, 9\}$$

Examine order of each element:

In $U(8)$, every non-identity element has order 2

In $U(10)$, this is not the case as $|3| \neq 2$

$\therefore U(8)$ and $U(10)$ have the same order but different structure regarding order of elements which violates property 7 of Theorem 4.2. $U(8)$ is not isomorphic to $U(10)$

5. $U(8) = \{1, 3, 5, 7\}$

$$U(12) = \{1, 5, 7, 11\}$$

Observe $|U(8)| = |U(12)|$.

Order of every non-identity element is 2 in both groups so group and elemental properties are validated.

Define Mapping $\phi: U(8) \rightarrow U(12)$

Since the sets have the same cardinality, the function is clearly injective and surjective as every element in $U(12)$ is an image of some element in $U(8)$.

O.P: show $\phi(xy \bmod 8) = \phi(x)\phi(y) \bmod 12$ for all x, y

Observe $\phi(1 \cdot x) = \phi(x)$ for any $x \in U(8)$ since $\phi(1) = 1$

$$\phi(2 \cdot 2) = \phi(1) = 1 \text{ and } \phi(2) \cdot \phi(2) = 5 \cdot 5 \bmod 12 = 1$$

$$\phi(2 \cdot 5) = \phi(7) = 11 \text{ and } \phi(2) \cdot \phi(5) = 5 \cdot 7 \bmod 12 = 11$$

$$\phi(3 \cdot 7) = \phi(5) = 7 \text{ and } \phi(3) \cdot \phi(7) = 5 \cdot 11 \bmod 12 = 7$$

$$\phi(5 \cdot 7) = \phi(3) = 5 \text{ and } \phi(5) \cdot \phi(7) = 7 \cdot 11 \bmod 12 = 5$$

Since these elements are self inverses and we've shown the operation is preserved for unique combinations, we say $U(8) \approx U(12)$

6 Let G, H, K be groups.

Reflexive: Define Mapping $\phi: G \rightarrow G$ as $\phi(g) = g$ for all $g \in G$.

$\phi(ab) = ab = \phi(a)\phi(b)$. This function is operation

preserving.

1-1: Let $\phi(a) = \phi(b)$. This means $a = b$ by definition.

This ϕ is injective

Onto: For every $c \in G$, there exists $a \in G$ such that

$\phi(a) = c$ since $a = c$. Thus ϕ is surjective

Symmetric: Let $G \approx H$ and $\phi: G \rightarrow H$ be an isomorphism.

Find ϕ^{-1} such that $\phi^{-1}: H \rightarrow G$

O.P: Since ϕ is operation preserving, for $h_1, h_2 \in H$, there exist $g_1, g_2 \in G$ such that $\phi(g_1) = h_1$ and $\phi(g_2) = h_2$

$$\begin{aligned} \text{Then } \phi^{-1}(h_1 h_2) &= \phi^{-1}(\phi(g_1)\phi(g_2)) = \phi^{-1}(\phi(g_1 g_2)) = g_1 g_2 \\ &= \phi^{-1}(h_1) \phi^{-1}(h_2) \end{aligned}$$

1-1 and onto: The inverse of a bijective function is also bijective

Transitive: Let $G = H$ and $H = K$ let $\phi: G \rightarrow H$

and $\psi: H \rightarrow K$ Find $\theta = \psi \circ \phi: G \rightarrow K$

$$\text{O.p. } \theta(ab) = \psi(\phi(ab)) = \psi(\phi(a)\phi(b)) = \psi(\phi(a))\psi(\phi(b)) \\ = \theta(a)\theta(b)$$

Thus θ is o.p.

1-1: Let $\theta(a) = \theta(b)$ This means $\psi(\phi(a)) = \psi(\phi(b))$

Since ψ is injective, $\phi(a) = \phi(b)$.

Since ϕ is injective, $a = b$

Thus θ is injective.

Onto: Let $k \in K$. Since ψ is surjective, there exists some $h \in H$ such that $\psi(h) = k$.

Since ϕ is surjective, there exists some $g \in G$ such that $\phi(g) = h$

Thus $\theta(g) = \psi(\phi(g)) = \psi(h) = k$ and θ is surjective.

$\therefore$ Since θ is reflexive, symmetric, and transitive, it is an equivalence relation.

11. Define Mapping: $\phi: G \rightarrow \bar{G}$ and $\phi(a) = \bar{a}$ and $\phi(b) = \bar{b}$

Since ϕ is an isomorphic mapping $\phi(a^3) = 3\phi(a)$

Thus $\phi(a^3) = 3\bar{a}$

$$\phi(a^3 b^{-2}) = \phi(a^3) + \phi(b^{-2}) = 3\bar{a} + -2\bar{b} = 3\bar{a} - 2\bar{b}$$

12. If α is an automorphism, G is Abelian:

Let $\alpha(g) = g^{-1}$ be an automorphism

$$\alpha(ab) = (ab)^{-1} \text{ and } \alpha(a)\alpha(b) = a^{-1}b^{-1}$$

Since α is o.p., $\alpha(ab) = \alpha(a)\alpha(b)$

$$(ab)^{-1} = a^{-1}b^{-1}$$

Taking inverses of both sides:

$$ab = (a^{-1}b^{-1})^{-1}$$

$$ab = ba$$

Thus G is Abelian.

If G is Abelian, α is an automorphism

Let G be Abelian, [show $\alpha(g) = g^{-1}$ is an automorphism]

O.p. $\alpha(ab) = (ab)^{-1}$ Since G is Abelian $(ab)^{-1} = (ba)^{-1}$

This means $b^{-1}a^{-1} = a^{-1}b^{-1}$

Thus $\alpha(ab) = \alpha(a)\alpha(b)$

1-1: Let $\alpha(a) = \alpha(b)$. This means $a^{-1} = b^{-1}$. Taking inverses of both sides $a = b$

Onto: For each $b \in G$ consider b^{-1} . Then $\alpha(b^{-1}) = (b^{-1})^{-1} = b$

Thus α is surjective

$\therefore \alpha$ is an automorphism if G is Abelian

31. Let $\phi: \langle a \rangle \rightarrow G$, $\gamma: \langle a \rangle \rightarrow G$ and $\phi(a) = \gamma(a)$

[show for every element $g = a^n$ in $\langle a \rangle$, $\phi(g) = \gamma(g)$]

Since ϕ and γ are isomorphisms, we know $\phi(a^n) = [\phi(a)]^n$
 $\gamma(a^n) = [\gamma(a)]^n$

Since $\phi(a) = \gamma(a)$ by definition:

$$\phi(a^n) = [\gamma(a)]^n = \gamma(a^n)$$

$\therefore$ since every element $g \in \langle a \rangle$ has the form a^n , we have shown $\phi(a^n) = \gamma(a^n) \therefore \phi = \gamma$

34. Let $K \leq G$ and $\phi: G \rightarrow \bar{G}$. [show $\phi(K) \leq \bar{G}$]

non-empty: Since $K \leq G$, $e \in K$. Thus, it follows $\phi(e) = \bar{e}$

Thus, identity element is in $\phi(K)$ and is non-empty

closure: Let $\bar{x}, \bar{y} \in \phi(K)$. This means there exists $x, y \in K$ such that $\phi(x) = \bar{x}$ and $\phi(y) = \bar{y}$. Since K is a subgroup, it is closed and $xy \in K$.

Since ϕ is o.p., $\phi(xy) = \phi(x)\phi(y) = \bar{x}\bar{y} \in \phi(K)$

Thus $\phi(K)$ is closed under the operation of $\bar{G}$

inverses: Let $\bar{x} \in \phi(K)$. We know there exists $x \in K$ such that $\phi(x) = \bar{x}$. Since K is a subgroup, $x^{-1} \in K$.

$$\phi(x^{-1}) = [\phi(x)]^{-1} = \bar{x}^{-1} \in \phi(K).$$

Thus $\phi(K)$ is closed under inverses

$\phi(K) \leq \bar{G}$ by 2.5.1

57 $\mathbb{Z}_{10}, S_5, D_{10}$

$\mathbb{Z}_{10} \ncong S_5$: Since $\mathbb{Z}_{10}$ and S_5 contradicts in Abelian group property, they cannot be isomorphic.

$\mathbb{Z}_{10} \ncong D_{10}$: Since $\mathbb{Z}_{10}$ is Abelian and D_{10} is not, they cannot be isomorphic.

$S_5 \ncong D_{10}$: S_5 have more complex subgroups like A_5 which aren't defined by geometric constraints.

Test 1 correction

1b. Product of even 2-cycles

2a. $|a^m| = n / \gcd(n, m)$

3j. $\mathbb{Z}_p$ just has 2 subgroups of order 1 and p with p^n , divisors are 1, p , p^n .

When $n=1$, there is clearly 1+1 subgroups

$\therefore \mathbb{Z}_{p^n}$ has $n+1$ subgroups

4a. identity: Since K is the set of all odd numbers, $0 \notin K$

closure: Let $a, b \in K$. This means $a=2n+1$; $b=2m+1$ where $m, n \in \mathbb{Z}$

$a+b = (2n+1) + (2m+1) = 2 + 2(n+m) = 2(n+m+1)$ is even and set is not closed

b. $|G|$ is a multiple of 7.

If there are only 3 subgroups and just one non trivial subgroup, order of G is 49. As divisors of 49 is $\{1, 7, 49\}$

5. Let $b = xax^{-1}$ and $|a| = n$. This means $a^n = e$ and $a^k \neq e$ on $0 < k < n$ interval

$$\begin{aligned} b^n &= (xax^{-1})^n \\ &= xax^{-1} \cdot xax^{-1} \cdot xax^{-1} \cdots \end{aligned}$$

$$= xa(x^{-1}x)a(x^{-1}x)ax^{-1} \dots$$

$$= xa^n x^{-1} = xe x^{-1} = e$$

Thus $b^n = e$

Assume for contradiction, there exist integer $k < n$ such that $b^k = e$

$$b^k = (xax^{-1})^k = xa^k x^{-1} = e$$

$$a^k = x^{-1}ex = e$$

This means $a^k = e$ which contradicts $|a| = n$.

$$\therefore |b| = n$$

Chapter 6

I. $\phi: n \rightarrow 2n$

16 $\phi_1(1) = 1$

$\phi_2(1) = 5$

$\{\phi_1, \phi_2\}$

III. Prove that $\text{Aut}(G)$ is a group:

Closure:

Let $\phi, \psi \in \text{Aut}(G)$. Since ϕ and ψ are bijective and o.p from G to G , their composition $\phi \circ \psi$ has same properties

- O.P:

$$(\phi \circ \psi)(xy) = \phi(\psi(xy)) = \phi(\psi(x)\psi(y)) = \phi(\psi(x))\phi(\psi(y))$$

$$= (\phi \circ \psi)(x)(\phi \circ \psi)(y)$$

- 1-1 & onto: Function is bijective

Thus $\phi \circ \psi \in \text{Aut}(G)$

Associativity:

$$(\phi \circ (\psi \circ \gamma))(x) = \phi(\psi(\gamma(x))) = (\phi \circ \psi) \circ \gamma(x)$$

$$\forall \phi, \psi, \gamma \in \text{Aut}(G)$$

Identity:

In $\text{Aut}(G)$ $e_G(x) = x$ for $\forall x \in G$

$$\phi \circ e_G = e_G \circ \phi = \phi$$

$\forall \phi \in \text{Aut}(G)$

Inverses:

Observe $\phi^{-1} \in \text{Aut}(G)$ when $\phi \in \text{Aut}(G)$

$$\phi \circ \phi^{-1} = \phi^{-1} \circ \phi = e_G$$

$\therefore \text{Aut}(G)$ is a group under function composition

Proof that $\text{Inn}(G)$ is a group:

closure:

Let $\phi_g, \phi_h \in \text{Inn}(G)$ where $g, h \in G$.

$$\begin{aligned} (\phi_g \circ \phi_h)(x) &= \phi_g(\phi_h(x)) = \phi_g(hxh^{-1}) = g(hxh^{-1})g^{-1} \\ &= (gh)x(gh)^{-1} \end{aligned}$$

Thus $\phi_g \circ \phi_h = \phi_{gh}$

Since $gh \in G$, $\phi_{gh} \in \text{Inn}(G)$, $\text{Inn}(G)$ is closed under composition

Associativity

Function composition is associative

Identity:

$$\phi_e(x) = exe^{-1} = x$$

Thus $\phi_e = e_G$ in $\text{Inn}(G)$

Inverses:

$$\phi_g \circ \phi_{g^{-1}} = \phi_{gg^{-1}} = \phi_e = e_G$$

$$\phi_{g^{-1}} \circ \phi_g = \phi_{g^{-1}g} = \phi_e = e_G$$

Thus, $\text{Inn}(G)$ is closed under taking inverse

* 21. Let $H = \{ \beta \in S_5 \mid \beta(1) = 1 \}$ and $K = \{ \beta \in S_5 \mid \beta(2) = 2 \}$

Since both H and K fix one element in S_5 , $H \cap K \cong S_4$

Define mapping: $\phi: H \rightarrow K$

Let $\gamma = (12)$

$$\gamma(1) = 2; \gamma(2) = 1; \gamma(3) = 3; \gamma(4) = 4; \gamma(5) = 5$$

$$(12)(13)(14)$$

$$(12)(13) = e$$

Observe $\gamma^{-1} = \gamma$

24 Let $H = \{x \in G \mid \phi(x) = x\}$ and $\phi: G \rightarrow G$ [show $H \leq G$]
non-empty:

Since ϕ is an automorphism $\phi(e) = e$

Thus $e \in H$

closure: Let $x, y \in H$. This means $\phi(x) = x$, $\phi(y) = y$

Since ϕ is o.p

$$\phi(xy) = \phi(x)\phi(y)$$

$$\phi(xy) = xy$$

Thus H is closed

Inverses: Let $x \in H$. This means $\phi(x) = x$

Since ϕ is o.p

$$\phi(x^{-1}) = [\phi(x)]^{-1}$$

$$\phi(x^{-1}) = x^{-1}$$

Thus H is closed under inverses

$\therefore H \leq G$ by 2SST

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28 Let $\phi: U(16) \rightarrow U(16)$

$$\phi(x) = x^3 \pmod{16}$$

o.p:

$$\phi(xy) = (xy)^3 \pmod{16} = x^3 y^3 \pmod{16}$$

$$\phi(x)\phi(y) = x^3 y^3 \pmod{16}$$

$$\text{Thus } \phi(xy) = \phi(x)\phi(y)$$

1-1: $U(16) = \{1, 3, 5, 7, 9, 11, 13, 15\}$

$$\phi(1) = 1^3 \pmod{16} = 1$$

$$\phi(3) = 3^3 \pmod{16} = 11$$

$$\phi(5) = 5^3 \pmod{16} = 13$$

$$\phi(7) = 7^3 \pmod{16} = 7$$

$$\phi(9) = 9^3 \pmod{16} = 9$$

$$\phi(11) = 11^3 \pmod{16} = 3$$

$$\phi(13) = 13^3 \pmod{16} = 5$$

$$\phi(15) = 15$$

Since $U(6)$ is finite and ϕ is clearly 1-1. ϕ is bijective
 $\therefore \phi$ is an automorphism of $U(6)$

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33 Let $\phi: G \rightarrow \bar{G}$ [show $\phi^{-1}: \bar{G} \rightarrow G$]
 1-1:

Since ϕ is bijective, every element in $\bar{G}$ maps to exactly one element in G . Thus ϕ^{-1} is injective

Onto: $\forall g \in G \exists \bar{g} \in \bar{G}$ such that $\phi(g) = \bar{g}$.

Thus $\forall g \in G, \phi^{-1}(\bar{g}) = g$ and ϕ^{-1} is surjective

O.p: Let $a, b \in \bar{G}$. Since ϕ is bijective $\exists x, y \in G$ such that

$$\phi(x) = a \quad \phi(y) = b$$

$$ab = \phi(x) \phi(y)$$

Also, since ϕ is o.p

$$\phi(xy) = \phi(x) \phi(y) = ab$$

$$\text{Now, } \phi^{-1}(ab) = \phi^{-1}(\phi(xy)) = xy$$

$$\text{Similarly, } \phi^{-1}(a) \phi^{-1}(b) = x \cdot y$$

$$\text{Thus, } \phi^{-1}(ab) = \phi^{-1}(a) \phi^{-1}(b) = xy \text{ and } \phi^{-1} \text{ is o.p}$$

$\therefore \phi^{-1}$ is an isomorphism from $\bar{G}$ to G

39. Disprove $(\mathbb{Z}, +) \cong (\mathbb{Q}, +)$

For isomorphisms to hold, group and elemental properties must match.

$(\mathbb{Q}, +)$ is the group of quotients and is "divisible".
 This means for any $q \in \mathbb{Q}$ and $n \in \mathbb{N}$, we can find $r \in \mathbb{Q}$ such that $nr = q$

In $(\mathbb{Z}, +)$ this is not the case as it is the group of integers

41) Let $x \in G$ and $\phi_g(x) = \phi_h(x) \quad \forall x \in G$ [shows $h'g \in Z(G)$]

This means $gxg^{-1} = hxh^{-1}$

$$gx = hxh'g$$

$$(h'g)x = x(h'g)$$

Thus $h'g$ commutes with $\forall x \in G$ and $h'g \in Z(G)$

52. If $\text{Inn}(G) = 1$, then it only contains the identity mapping such that $\phi_g(x) = x$

If and only if G is Abelian